

# Admissibility of Digital Evidence in Court: Legal Standards and Forensic Practice

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**Abstract**— The admissibility of digital evidence is one of the most important issues in recent investigations, because of the growing reliance of digital platforms and devices. This study introduces an enhanced model for checking the admissibility of digital evidence. The E-DEF Enhanced Digital Evidence Admissibility Framework that improves Antwi-Boasiako and Venter structure by adding more phases based on established international standards, including ISO/IEC 27037 and 27041, NIST SP 800-86 and 800-101, SWGDE guidelines, is designed to fill the gaps in Harvey's analysis.

**Keywords**—digital evidence, forensic standards, admissibility, E-DEAF

## I. INTRODUCTION

Digital proof has turned out to be one of the most crucial elements in today's investigations taking part in the influencing criminal, civil, and cybersecurity lawsuits. Along with the increase of technology, the digital artifacts, like mobile data, social media contents, and electronic logs, have also been increased in number and complexity to a great extent. Yet, digital evidence remains to be very delicate, easy to change, and in most cases, not clear on who it belongs to which makes its correct handling and acceptance a big problem for both the investigators and the courts. This, in turn, makes the need for very high standards and regulations to be in place at all times to guarantee the security, genuineness, and legal reliability of the evidence during the whole investigation.

The international standardization agency's ISO/IEC 27037 has laid down the procedure for identifying, collecting, acquiring, and preserving digital evidence in a way that it can be used in court as it is a forensic evidence. The Daubert standard, on the other hand, creates scientific criteria, including testability, peer review, and known error rates, to judge the reliability of forensic methods and expert testimonies. Besides, NIST and SWGDE provide additional assistance by highlighting the importance of tool validation,

metadata preservation, and proper documentation while also warning against the use of non-native formats, such as screenshots, that lack metadata and are easily manipulated. Even though these regulations greatly assist the forensic practice, the literature that is written on this matter is still pointing out alike problems of digital data being fragile, difficulties in social media identity verification, and lack of

a common framework which bridges the gap between legal and technical standards.

Antwi-Boasiako and Venter's model and other previous academic works highlight a three-step process that includes legal authorization, technical evaluation, and judicial adjudication. However, while this model is good, it is still missing some elements that are needed to deal with the current problems, for instance, enforcing native formats, digital identity verification, assessing the reliability of the platform, and having a structured evaluation by the judiciary. Moreover, Harvey's critique points out that social media evidence is always deprived of metadata, cannot be traced back to a certain account, and is generally very prone to manipulation, thus making the traditional evidence rules a very challenging issue.

With all these limitations, the present research puts forward a new and improved framework that is called E-DEAF (Enhanced Digital Evidence Admissibility Framework). The five phases are the ones that E-DEAF developed by taking the first three phases model from the international standards of ISO/IEC 27037, ISO/IEC 27041, Daubert, NIST guidelines, and SWGDE recommendations. Among other things, it has introduced new tasks of pre-evidence validation, authenticity verification, and a quantitative judicial scoring system. Hence, E-DEAF becomes a well-rounded, contemporary model that equips the admissibility of digital evidence by marrying up legal requirements, technical integrity, digital identity verification, and standardized evaluation criteria through its expansions.

## II. PROBLEM STATEMENT

Despite the availability of international guidelines (ISO/IEC 27037, NIST, SWGDE) and legal admissibility tests (Daubert, FRE), these models are still not enough to address the issues related to digital-identity verification, platform reliability assessment, metadata preservation, and the evaluation of non-native evidence formats such as screenshots. These not addressed problems leading to unsteady judicial outcomes. Therefore, a comprehensive framework should be created to ensure the integrity, authenticity, and admissibility of modern digital evidence.

## III. BACKGROUND

To ensure authenticity and integrity in the investigative process, digital evidence handling requires strict procedures for acquisition, preservation. One of the most influential frameworks in prior work is a model for the admissibility of digital evidence by Antwi-Boasiako and Venter[15], which organizes the process into three main phases: legal authorization, technical examination, and judicial determination. Harvey's work further expands on the broader challenges associated with digital evidence -highlights issues[16], such as the inherent fragility of digital data, the ease with which electronic information can be altered, the lack of reliable attribution in many digital environments, and the particular difficulties posed by social media evidence, including unverifiable account identities, incomplete metadata, and the prevalence of easily manipulated screenshots. Despite these contributions, the literature lacks of unified and comprehensive framework that integrates international and even national standards, addresses the modern complexities of digital identity verification, enforces native-format requirements to ensure evidentiary integrity, and provides structured, measurable criteria for judicial assessment. The enhanced E-DEAF framework proposed in this study directly responds to these gaps by combining legal, technical, and authenticity- centered mechanisms into a coherent and modern admissibility model.

## IV. LITERATURE REVIEW

### A. ISO/IEC 27037: Guidelines for Handling Digital Evidence

The ISO/IEC 27037 standard plays the role of an international standard and it is the first one that describes the best practices for the processes of identification, collection, acquiring, and preserving digital evidence. The core of the standard is the right conservation of the evidence in its original format, the keeping of the metadata, the uninterrupted chain of custody, and the use of reliable forensic tools. Moreover, the standard demands the alteration of the data to be kept at a minimum, and every stage of the evidence-handling process should be thoroughly documented.

Strength: Offers a solid technical way to maintain the integrity of evidence.

Limitation: Mechanisms for digital identity verification or assessments of social media platforms' reliability are not included.

Gap: Modern challenges concerning digital identity or non-native evidence like screenshots are not covered.[1]

### B. Daubert Standard: Scientific Admissibility Criteria

The Daubert Standard sets out guidelines for the judicial acceptance of science and technology evidence. The four main cornerstones—testability, peer review, known error rate, and general acceptance—are the basis for judgment of forensic tools and methods.

Strength: Bridges the scientific norms and the legal standards so that forensic techniques can be trusted.

Limitations: It does not specify the technical methods for the gathering or the processing of the evidence.

Gap: It does not tackle the issues of digital identity nor does it take into account the peculiarity of social media evidence.[4],[13]

### C. NIST Forensic Guidelines

NIST periodically publishes influential digital forensic guidelines which are listed below:

SP 800-86: A guide for incorporating forensic processes into the investigations.

SP 800-101: Guidelines for the forensic examination of mobile devices.

CFTT: Guidelines for determining the reliability of forensic instruments.

These guidelines mainly deal with the issues of tool validation, hashing, error rate calculation, repeatability, and traceability.

Strength: Provides very precise technical standards and also imposes strict measures for the validation of forensic tools.

Limitation: Only technical procedures are emphasized; a comprehensive judicial admissibility framework is not provided.

Gap: User verification and platform reliability evaluation are not considered as part of the process.[2],[5],[9]

### D. SWGDE Standards

The Scientific Working Group on Digital Evidence delivers the best practices for digital forensics, which are very often referenced. It is important to mention that SWGDE

specifically emphasizes “Screenshots are NOT forensic evidence” as they do not have metadata and are very prone to manipulation.

**Strength:** It gives a precise direction that the formats of evidence which are weak, like screenshots, should not be allowed.

**Limitation:** It does not provide the whole legal framework for the evidence admissibility.

**Gap:** It does not talk about the judicial evaluation processes or digital identity verification.[6],[10]

#### E. A Model for Digital Evidence Admissibility Assessment

The discussions around digital evidence have pointed out that the biggest hurdles in guaranteeing the presentation of digital evidence in courts being admissible and reliable are the quick pace of technology and the international character of cybercrimes. On the one hand, Antwi-Boasiako and Venter underline the fact that digital forensics has become a crucial part of modern investigations, yet it is still devoid of a comprehensive, uniform framework that binds the technical and the legal requirements for admissibility across different jurisdictions. In the past, forensic practices have been developed in a fragmented manner, and thus, they have resulted in the emergence of diverse models, inconsistency in procedures, and differences in the treatment of evidence. Among the past attempts to reconcile the digital forensic operations there were the models like DFRWS, Reith et al.’s abstract forensic model, guidelines of ACPO, and ISO/IEC standards that were some of the notable ones. Although these initiatives improved the consistency in the areas of identification, acquisition, and preservation, they still did not cover admitting the evidence directly. Research has been revealing that valid digital evidence is to be relevant, authentic, reliable, and legally collected, but the current standards do not make the legal requirements assessment and the technical procedures connection in forensic investigations clearly. Hence, the courts frequently encounter the problem of whether the evidence was correct in obtaining, and processing with validated forensic tools, preserved with proper chain of custody, and handled by qualified experts.

A different topic that appears in the literature is the susceptibility and instability of digital evidence. According to the writers, metadata can be changed or integrity can be compromised and thus, very careful technical assurance is required. The verification process consists of utilizing validated tools, following correct laboratory procedures, employing hashing, using write blockers, and creating reproducible forensic reports. Furthermore, legal scholars have been emphasizing the need for warrants, privacy

protections, authenticity assessments, and reliability criteria similar to those introduced in the Daubert ruling.

A coalescence of studies indicates a distinct demand for a synchronized techno-legal framework that unifies the legal requirements and the scientific forensic requirements. Antwi-Boasiako and Venter fill this void by putting forward a single model consisting of three stages—legal assessment, technical consideration, and final evidential determination—which is intended to allow the digital evidence to be evaluated for admissibility and weight consistently across different jurisdictions. Their research lends support to the developing agreement that digital forensics needs integrated standards that will not only support cross-border investigations but also adapt to the inevitable technological changes.

**Strength:** Underlines the importance of a comprehensive techno-legal framework and at the same time defines properly the key forensic requirements such as validated tools, maintenance of the chain of custody, and compliance with legal standards like authenticity and reliability.

**Limitation:** Current models and standards (DFRWS, ACPO, ISO/IEC) mainly address technical procedures and do not entirely connect them to the judicial admissibility criteria applicable in different jurisdictions.

**Gap:** Absence of a globally recognized framework merging legal evaluation with technical forensic processes, particularly in the case of transnational cybercrimes and emerging technologies. [15]

#### F. Digital Evidence Admissibility Some Issue

Research concerning digital evidence continually emphasizes the point that digital data is a different category of evidence from the traditional counterpoint of analogue evidence mainly because it is volatile, can easily be modified, and along with such is also dependent on very intricate technological systems for its interpretation. Harvey states that the courts are “information hubs,” and the amount of reliability of the verdicts is determined by the quality and the origin of the digital evidence that is presented in court. In contrast to the immovable paper documents, digital data is unstable by its nature and thus must go through very rigorous authentication processes to ascertain that it is accurate and its integrity is intact. A widespread issue, as pointed out by the researchers, is the fact that the legal system relies on screenshots, printouts, and photos of screens. These evidences not only lack the metadata and contextual information which are essential for verification but also often result in questioning the reliability and provenance of the evidence.

The literature conveys that the procedure of digital evidence authentication is no longer solely about providing the content along with the name of the author; instead it is all about the IP address connections, writing style harmony, the use of native digital formats, and by showing the access of devices and authorship of the document. The hurdles that are very frequently mentioned in the discussions are identity

impersonation, digital trickery, incomplete records, and also the problems of connecting content from anonymous or fake accounts to certain individuals. Among the issues of social media evidence, platform structure, creating accounts, setting one's privacy level, and the possibility of third-party access are the factors that severely complicate attribution and thus are the issues that are most visible. The scholars observe that courts make erroneous assumptions very often, for instance, posting something on Facebook is interpreted as the user being willing to disseminate the information broadly, thus illustrating a rift between the legal rationale and the technological reality.

Another important aspect of the discussion is the machine-generated evidence. The current evidence law, e.g. section 137 of the Evidence Act 2006, assumes the accuracy of a machine or technical procedure only if it is of a kind that "ordinarily" performs reliably. Publications indicate that this assumption developed from the "notorious instrument" doctrine, which involved such devices as computers, GPS units, and breathalyzers gradually gaining acceptance after long periods of demonstrated reliability. Nevertheless, courts still demand verification when dealing with unrecognized software, new forensic tools, or untried systems, and a failure to prove the ordinary operation may lead to the evidence being ruled out. Researchers also point out that the digital chain of custody has to be treated with the same strictness as physical evidence and made sure to be documented at every step from collection to presentation.

In short, the literature indicates a gradual agreement among the stakeholders that the lawyers should be acquainted with the technological processes as digital information will be the main source of modern litigation. Proper authentication, native-format preservation, technical verification, and platform-specific analysis are the measures that must be taken to prevent "fake evidence" from getting into the courtroom and to make sure that digital material is of the quality required for admissibility.

**Strength:** It distinctly points out the volatility of digital evidence that is unique and stresses the need of the stringent authentication methods such as the analysis of metadata, tracing the IP and verifying in-native format.

**Limitation:** The paper mainly discusses the theory and puts all its emphasis on the issues that have already been identified, such as the use of screenshots, without actually providing any measures of standard procedural solutions for the courts.

**Gap:** There are no empirical studies on how frequently digital evidence is accepted or rejected in actual cases and the paper also does not cover new challenges like the AI-generated content or deepfake editing. [16]

## V. INTERNATIONAL STANDARDS(DAUBERT,ISO/IEC 27037)

In U.S. courts, the digital evidence admit ability is under the Daubert standard, which permit judges to take the role of gatekeepers to make sure that the expert testimony is reliable and relevant. The standard sets forth a uniform procedure for the assessment of forensic methods, devices, and findings by prioritizing scientific validity, empirical testing, error rates, and the professional community's broad acceptance using a very structured and rigorous evaluation process. Those organizations that follow the Daubert guidelines can offer their operations, assessments, and expert witness testimonies as supporting evidence that is scientifically valid, reproducible, and defensible, thus creating a legal reliability and courtroom credibility that is built on a strong foundation.

The Daubert standard was born out of a trilogy of Supreme Court cases: *Daubert v. Merrell Dow Pharmaceuticals*, *General Electric Co. v. Joiner*, and *Kumho Tire Co. v. Carmichael*. The rulings of these three cases together clarified that judges act as gatekeepers who make sure the expert opinions are based on scientifically valid methods and have been correctly applied to the particular facts of the case. Under this standard, electronic evidence is assessed in such a manner that the forensic techniques, analytical instruments, and expert thinking are all subjected to objective criteria for reliability, and if they are not, the examinee's skills and claims may not hold up.

At the heart of Daubert is the combination of two-pronged test of relevance and reliability. First, relevance calls for expert testimony to be a help to the factfinder in the process of understanding or deciding a material issue. Then, reliability requires that the product of such a testimony be a consistent, verifiable, and defensible methodological process. This is true in the case of digital forensics too, where it applies across the board from the very beginning to the very end of evidence handling, that is, through disk imaging, metadata extraction, network capture, and log analysis. The courts are going to take a digital forensic evidence case through the Daubert criteria, one after the other, asking about empirically testable techniques, the occurrence of peer review or publication, the error rates of known or potential kinds, the presence of standards and controls, and the acceptance within the forensic community that the degree of evidence has reached.

The standard is also a pointer to the importance of having a sound method. Scientific knowledge must be interpreted in such a way that it is possible to demonstrate conclusively that the results are the truth. As per Daubert, this implies that scientific methods are used, such as formulating a hypothesis, conducting experiments, and proving the opposite of the

hypothesis. In digital evidence, this means using approved tools and methods, documenting every detail of the process, and making the whole process easily reproducible. Tests that are specifically designed for litigation or do not involve any research at all are treated more sceptically whereas methods with peer-reviewed validation, reproducible results, and community recognition are considered to be stronger ones.

Judge-gated gatekeeping is a basic characteristic of Daubert. The courts check the qualifications of the experts, which means that the knowledge, skills, experience, training, and education required to support the testimony are evaluated to be sufficient. Nevertheless, just having the expertise is not enough; opinions should be formed by utilizing sound principles and methods. Proper chain-of-custody documentation validated acquisition and analysis tools, and evidence-based rather than speculation-based justifications for interpretations would be the digital forensic issues of concern. Besides, the judge is curious to discover if the expert has excluded other possible explanations and has not gone beyond data to draw conclusions without first taking the right steps.

The criteria set forth by the Court provide a clear and practical means to assess the reliability of digital evidence:

**Empirical Testing** – Procedures like hashing, write-blocked imaging, or memory extraction should not only be available for testing but also be able to produce the same results when performed in different settings. **Peer Review and Publication** – When the methods are published in forensic journals, standards, or protocols recognized by the industry, this indicates that they are credible.

**Known or Potential Error Rate** – The reliability of the methods can be measured through the use of error metrics, such as the probabilities of hash collisions or the rates of errors of the extraction tools, which will reveal the mistakes that could be made in the operations.

**Standards and Controls** – The adoption of ISO standards (e.g., ISO/IEC 27037, 27041, 27042, 17025) assures a regulated yet delineated supporting path throughout the methods.

**General Acceptance** – The acceptance of the techniques by the forensic community and their incorporation into the industry best practices are undoubtedly signs of the credibility of the methods.

Daubert stipulates that the use of expert testimony must be tied to the facts. This includes the proper application of the techniques, correct reading of the results, and reasoning that is in proportion to the data collected. Testimony might be rejected by the court if there is huge disparity between the foundational data and the expert opinion, if the choice of methods was dictated by litigation purposes, or if alternative explanations were ruled out.

In the arena of electronic evidence, the courts apply Daubert along with specific procedural rules such as chain-of-custody principle, authentication requirements of FRE 901, and best

evidence rule. These rules connect, verify, and represent digital evidence as the same as the original from the time of its acquiring up to its presenting. The Daubert Framework may be employed within the organization to establish structured workflows which would cover tool validation, examiner qualifications, documentation, analysis, and reporting. This not only makes evidence admissible in court but also minimizes the risk of evidence being ruled inadmissible, and aligns forensic practices with acknowledged scientific standards.

The primary focus of Daubert is to assess if the expert testimony meets the criteria of being both relevant and reliable. To be reliable it has to be supported by a consistent, verifiable, and sound methodology which is properly applied to the particular evidence. In the field of digital forensics this would mean passing the tools through validation, making thorough documentation of the steps taken for acquisition and analysis, and ensuring reproducibility. Also, the judges will take into account whether the experts have knowingly disregarded the counter-arguments and have not made baseless claims from data to opinion. The methods used only for litigation are usually weighed more carefully than the established ones that have been peer-reviewed.

According to Daubert, the practical recommendations include: the testing of procedures such as hashing and imaging through empirical methods; peer review or publication as a means of demonstrating credibility; the error rates being quantified; the conformity to standards and controls; and the recognition by the professional community. These factors act as a structured framework that organizations can rely on to reinforce the credibility of their analysis and expert testimony.

In the end, the Daubert standard facilitates the use of proper methods by credible experts, of the digital evidence that has been subjected to collection, analysis, and presentation; in fact, it is based on the case factors being pertinent and the application of strict methodologies, along with the aspects of openness and responsibility. When organizations align their forensic practices with the Daubert factors, they render their digital evidence and expert testimony immune from court scrutiny, thus maintaining their credibility and assisting in the delivery of trustworthy legal outcomes.[4]

Digital evidence is subject to strict handling procedures for it to remain reliable, admissible and scientifically defensible. The lifecycle from identifying the piece of evidence to its use in a trial or investigative context is supported by a set of specialized ISO/IEC standards. These standards help organizations keep their integrity, validate forensic tools, ensure unbiased analysis, and protect evidence during the whole process going from cybersecurity incidents to lab

operations. By utilizing the above-mentioned framework, detectives and digital forensics teams can create consistent procedures that are in accordance with both technical and legal requirements.

Handling of digital evidence is governed by a limited number of ISO/IEC standards that when put together create the complete lifecycle: starting from identifying evidence, validating forensic tools, analysing data, responding to incidents, and ensuring the integrity of evidence for legal proceedings. The ISO/IEC 27037 is the foundation, as it specifies the way how digital evidence is to be identified, preserved, and acquired in such a manner that its integrity is maintained right from the time it is discovered. The standard lays stress on the use of forensic sound methodologies, ensuring the least possible alteration, and maintaining an unbroken chain of custody. Any organization that adheres to this standard will later be able to develop checklists for workflows of evidence acquisition, procedures for imaging data storage, and documentation of custody.

In the ISO/IEC 27041, the methods and tools used for assurance of evidence acceptance are not only reliable but also scientifically valid. It directly connects to the Daubert standard and other similar court requirements because it delineates the establishment of forensic tools, the backing of methodology, and the guarantee of repeatability. The ISO/IEC 27041 can help in formulating checklists for the validation of imaging software, the proper calibration of tools, and the defense of evidence extraction procedures.

During the examination phase, ISO/IEC 27042 takes over the process. It emphasizes the importance of interpreting digital evidence in a neutral and objective way, thus enabling the examiners to communicate their findings in a clear and consistent manner. The standard enables the formulation of systematic analysis procedures, documentation rules, review steps, and interpretation guidelines, which in turn lead to developing checklists for examination notes, reports, and result verifications.

In the case of incidents where evidence is present in real-time operational settings, the ISO/IEC 27043 standard lays down the principles guiding incident investigation. The standard merges forensic processes with cybersecurity incident response, thus securing the acceptance of the evidence collected during the attacks. This standard offers the possibility for organizations to later develop checklists for methods of handling incidents, incident containment, logging requirements, evidence preservation during IR, and coordination between responders and forensic teams.

ISO/IEC 27050 broadens the coverage to electronic discovery, which is the major part of legal proceedings that require the handling of vast amounts of digital documents. It becomes clearer how to identify, process, and approve the

trial evidence. It also helps in the establishment of procedures for the collection of emails, logs, and documents which will be defensible and legally acceptable.

Besides these fundamental digital-evidence standards, ISO/IEC 17025 has a pivotal role in providing that forensic labs are technically proficient and have reliable quality control practices. Accreditation based on this standard adds to the trustworthiness of the evidence that is presented in the court. It opens the door for the future checklists including lab practices, quality redemption, tools calibration, and competency of examiners.

Alongside the supporting standards such as ISO/IEC 27001, which ensures that digital evidence is protected even after the collection, by allowing the implementation of evidence storage systems, access control, and logging as well as data protection policies that maintain the integrity and confidentiality of the evidence. ISO/IEC 30121 further enhances digital forensic operations by introducing governance and risk management to mitigate the risks of tampering, misinterpretation, or procedural failure.

ISO/IEC standards provide both operational and technical advice that can be relied on and argued in court for the digital evidence that they have been able to keep in good condition. The main standards propose organized procedures for every step of the evidence handling process – identification, acquisition, validation, analysis, incident response, lab operations, and secure storage. For instance, ISO 27041 is about the validation of methods, ISO 27042 is about the unbiased analysis and documentation, and ISO 17025 is about the accreditation of labs that can prove their technical competence and reproducibility. Besides, ISO 27001 and 30121 are collaborating in the area of establishing the regulatory framework, risk management, and evidence protection.

The existing standards create a networked framework that instantly certifies the reliability of the evidence being through every stage of the process. These standards encompass the areas of identification, preservation, validity, analysis, incident handling, laboratory quality, and secure storage—enabling any given organization to translate such analysis into concrete operational checklists for digital forensics, chain-of-custody processes, and legal admissibility.[1],[7],[12]

## VI. ENHANCED MODEL (E-DEAF)

The enhanced model is developing based on Antwi-Boasiako&venter model and Harvey's paper suggestions. We developed E-DEAF, which is the ENHANCED DIGITAL EVIDENCE ADMISSIBILITY FRAMEWORK, using international standards such as ISO/IEC 27037, ISO/IEC 27041, Daubert, SWGDE, and NIST forensic guidelines. The

original model contained three phases only, but when Harvey's suggestions addressed the E-DEAF model contained five phases, as follows:

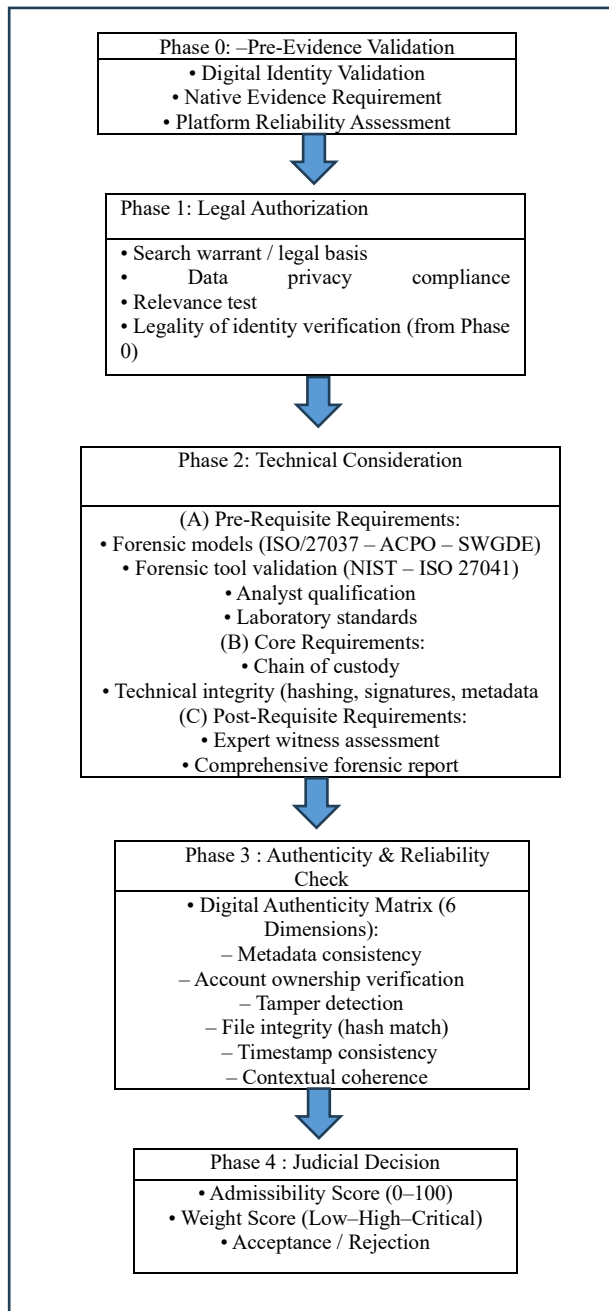


Figure (1)

The flowchart at figure (1) shows E-DEAF . explaining of each phase as following:

#### A. phase 0 ; pre-evidence validation

| Requirement                     | Verification Questions                                                                                                                                                               | Reference                                     |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Digital Identity Verification   | <ul style="list-style-type: none"> <li>Is the account linked to the suspect through IP logs, Device ID, MAC, login history?</li> <li>Is stylometric profiling consistent?</li> </ul> | NIST SP 800-86; FRE 901(b)(4)                 |
| Native Format Requirement       | <ul style="list-style-type: none"> <li>Is the evidence presented in its original native form (no screenshots)?</li> <li>Are metadata fields intact?</li> </ul>                       | SO/IEC 27037; NIST 800-101                    |
| Platform Reliability Assessment | <ul style="list-style-type: none"> <li>Does the platform retain logs?</li> <li>Is the platform vulnerable to manipulation?</li> </ul>                                                | ISO/IEC 27041; SWGD E Social Media Guidelines |

Table (1)

#### B. phase 1: legal authorization

| Requirement               | Verification Questions                                                                                                                          | Reference                |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Lawful Collection         | <ul style="list-style-type: none"> <li>Was a valid warrant issued?</li> <li>Was collection conducted without violating privacy laws?</li> </ul> | GDPR Principles; FRE 401 |
| Relevance and Scope       | <ul style="list-style-type: none"> <li>Is the evidence directly related to the investigation?</li> </ul>                                        | FRE 402                  |
| Authorization Audit Trail | <ul style="list-style-type: none"> <li>Is there documentation verifying lawful acquisition?</li> </ul>                                          | ISO/IEC 27037 Clause 6   |

Table (2)

#### C. Phase 2 — Technical Examination

##### 1) Pre-Requisite Requirements

| Requirement        | Verification Questions                                                                                                                    | Reference                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Validated Tools    | <ul style="list-style-type: none"> <li>Was a tested and certified forensic tool used?</li> <li>Is the tool version documented?</li> </ul> | Daubert (Error Rate); NIST CFTT |
| Competent Examiner | <ul style="list-style-type: none"> <li>Is the examiner trained and certified?</li> </ul>                                                  | ISO/IEC 17025                   |

Table (3)

### 2) Core requirements

| Requirement                | Verification Questions                                                                                                 | Reference                       |
|----------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Chain of Custody (3-Layer) | <ul style="list-style-type: none"> <li>Is there complete documentation of collection → transfer → analysis?</li> </ul> | ISO/IEC 27037; ACPO Principle 3 |
| Integrity Verification     | <ul style="list-style-type: none"> <li>Are hash values (MD5/SHA-256) recorded before and after analysis?</li> </ul>    | SWGDE Integrity Guidelines      |
| Forensic Imaging           | <ul style="list-style-type: none"> <li>Was a bit-for-bit forensic image created?</li> </ul>                            | NIST 800-101                    |

Table (4)

### 3) Post-requisite requirements

| Requirement                   | Verification Questions                                                                                           | Reference     |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|---------------|
| Forensic Report Completeness  | <ul style="list-style-type: none"> <li>Does the report describe tools, steps, settings, and findings?</li> </ul> | ISO/IEC 27041 |
| Expert Testimony Preparedness | <ul style="list-style-type: none"> <li>Can the expert defend methods under Daubert scrutiny?</li> </ul>          | FRE 702       |

Table (5)

### D. Phase 3 — Authenticity & Reliability check

| Requirement             | Verification Questions                                                                                | Reference                 |
|-------------------------|-------------------------------------------------------------------------------------------------------|---------------------------|
| Metadata Consistency    | <ul style="list-style-type: none"> <li>Timestamps, file origin, and authorship consistent?</li> </ul> | ISO/IEC 27037             |
| Account Ownership Proof | <ul style="list-style-type: none"> <li>Identity linked conclusively to suspect?</li> </ul>            | FRE 901                   |
| Tamper Detection        | <ul style="list-style-type: none"> <li>Any indication of editing or manipulation?</li> </ul>          | Daubert (Testability)     |
| Hash Matching           | <ul style="list-style-type: none"> <li>Do hash values match across all stages?</li> </ul>             | SWGDE Integrity Checklist |
| Timestamp Logic         | <ul style="list-style-type: none"> <li>Are timing sequences coherent with case facts?</li> </ul>      | NIST SP 800-86            |
| Contextual Reliability  | <ul style="list-style-type: none"> <li>Does evidence logically correlate with other data?</li> </ul>  | FRE 403                   |

Table (6)

### E. Phase4: Judicial Scoring System

| Category            | Weight |
|---------------------|--------|
| Legal Authorization | 20     |
| Integrity           | 20     |
| Authenticity        | 20     |
| Reliability         | 20     |
| Relevance           | 20     |

Table (7)

The evidence is evaluated on 100 point scale:

Admissible ≥60

Not admissible <60

This scoring method strengthens judicial transparency and aligns with ISO 27041 evidence evaluation guidelines[7].

## VII. COMPARATIVE ANALYSIS

| Element                            | Original Model | Enhanced Model (E-DEAF)                                       | Difference / Added Feature                 | Supporting International Reference                   |
|------------------------------------|----------------|---------------------------------------------------------------|--------------------------------------------|------------------------------------------------------|
| Number of Phases                   | 3 phases       | 5 phases                                                      | Added Phase 0 and Phase 3                  | ISO/IEC 27037 [1]                                    |
| Digital Identity Verification      | Not included   | Full identity validation (IP / Device ID / Logs / Stylometry) | Solves issue of fake accounts              | NIST SP 800-86 [2], FRE 901(b)(4) [3], Daubert [4]   |
| Native-Format Evidence Requirement | Not addressed  | Mandatory native format; rejects screenshots                  | Solves missing metadata                    | ISO/IEC 27037 [1], NIST SP 800-101 [5]               |
| Automatic Screenshot Rejection     | Not addressed  | Screenshots rejected as non-forensic                          | Ensures integrity and authenticity         | SWGDE Social Media Guidelines [6], ISO/IEC 27037 [1] |
| Platform Reliability Assessment    | Not included   | Platform evaluation (logs, tampering risk, publishing system) | Addresses social media challenges (Harvey) | ISO/IEC 27041 [7], NIST SP 800-145 [8]               |



|                                            |                     |                                                         |                                 |                                                        |
|--------------------------------------------|---------------------|---------------------------------------------------------|---------------------------------|--------------------------------------------------------|
| <b>Tool Documentation &amp; Error Rate</b> | Partially mentioned | Full tool documentation: version, error rate, usage log | Aligns with Daubert reliability | Daubert [4], NIST CFTT [9], SWGDE Tool Validation [10] |
| <b>Chain of Custody</b>                    | Included            | Expanded to 3 layers with timestamps                    | Matches ISO evidence handling   | ISO/IEC 27037 [1], ACPO Guide [11]                     |
| <b>Expert Reporting</b>                    | Included            | Added Expert Validation Matrix                          | Supports expert assessment      | FRE 702 [13], ISO/IEC 17025 [12]                       |
| <b>Judicial Decision</b>                   | Very general        | Structured scoring (Admissibility + Weight Score)       | Measurable judicial evaluation  | ISO/IEC 27041 [7], FRE 403 [14]                        |
| <b>Practical Applicability</b>             | Academic            | Practical and applicable in Saudi investigations        | Operational forensic framework  | ISO/IEC 27037 [1], NIST Guidelines [2][5]              |

Table (8)

### 3) Evidence Examination

The dataset included the following:

1. A disk image (**baseCFReDS01.dd**) containing hidden UTF-16BE text.
2. A ZIP archive (**russian-utf-16.zip**) with Russian and English UTF-16BE sample files.

The disk image was analyzed using FTK Imager, which enabled Unicode string searching and decoding. The provided sample files were used to verify recovered strings and ensure proper UTF-16BE interpretation, which stores each character in two bytes and requires careful decoding to avoid misinterpretation.

### 4) Tools and Technical Approach

FTK Imager (v4.2.0.13) was the primary forensic tool used. It enabled the examiner to:

- Load and inspect the disk image
- Open and analyze ZIP archives
- Perform Unicode (UTF-16) string searches
- Decode Russian text using Hex View and Text View

These capabilities facilitated accurate recovery of multilingual text embedded inside the forensic image.

### 5) Methodological Analysis

The investigation followed a systematic methodology:

1. Loading the disk image as evidence.
2. Reviewing the UTF-16 sample files for reference.
3. Running Unicode string searches (UTF-16 mode enabled) to detect headings.
4. Extracting decoded results from Text View.
5. Reconstructing the full menu by organizing the UTF-16BE strings sequentially.

### 6) Findings

All eight Russian menu sections were successfully recovered and reconstructed:

1. ЗАКУСКИ
2. СУП
3. БЛИНЫ
4. ПИРОЖКИ И ПЕЛМЕНИ
5. МЯСО И РЫБА
6. СЫР И МОЛОЧНЫЕ
7. НАПИТКИ
8. СЛАДКОЕ

The recovered strings were complete, accurate, and properly decoded.

## VIII. CASE STUDIES

### A. Case 1: Russian UTF-16BE Forensic Dataset

#### 1) Case Background

This forensic case involves a controlled dataset provided for digital-forensics training, focusing on the recovery of Russian UTF-16BE encoded text hidden within a disk image. The simulated scenario represents the theft of a Russian restaurant menu, whose content was concealed inside the image using Unicode encoding. The case is academically significant because it highlights challenges related to multilingual text recovery and encoding-aware forensic techniques.

#### 2) Incident Description

According to the scenario, two individuals—Boris and Natasha—stole a Russian Tea Room menu and embedded its contents inside a disk image. The stolen menu was divided into eight sections, all encoded in UTF-16BE. The examiner's task was to locate, decode, and reconstruct these sections using tools capable of interpreting Unicode-encoded data.

## 7) *Discussion*

This case demonstrates the limitations of ASCII-based search techniques in forensic analysis. Because UTF-16BE uses two bytes per character, ASCII searches cannot detect the hidden Russian text. The investigation highlights the importance of Unicode-aware forensic workflows and shows that improper handling of Unicode datasets can lead to incomplete or incorrect results. FTK Imager proved effective for analyzing multilingual encoded data.

## 8) *Summary of Insights*

This case illustrates:

- The importance of Unicode-aware forensic techniques
- The necessity of tools supporting UTF-16BE decoding
- How multilingual datasets can be embedded inside forensic disk images
- The value of systematic reconstruction of encoded text

Overall, the forensic process successfully recovered and reconstructed the hidden Russian menu.

## B. *Case 2: Saudi Digital Extortion Case*

### 1) *Case Background*

This case, documented in the Kingdom of Saudi Arabia under Case No. 3492328, involves a defendant who engaged in online extortion, defamation, and threats against a female victim through social-media platforms. The defendant repeatedly targeted the victim by name and identity. The case holds legal importance due to the presence of aggravating circumstances, including a prior extortion conviction.

### 2) *Incident Description*

The defendant published photos of the victim online, attached to her real identity, and used these posts to threaten her. Written messages contained explicit threats aimed at intimidating and harming the victim both emotionally and socially. These acts fall under criminalized behavior in Saudi cybercrime law. The Criminal Court of Taif convicted the defendant, and the Court of Appeal in Makkah reviewed the case multiple times, each time requesting a harsher sentence due to repeated offenses.

### 3) *Digital Evidence Examination*

The evidence collected included:

1. Verification of the defendant's personal and official social-media account.
2. Photos of the victim posted publicly.
3. Screenshots of explicit written threats.
4. Documentation of a prior extortion case.
5. A recorded confession by the defendant.

Forensic analysis confirmed that the posts and messages originated from the defendant's verified social-media account. The combination of textual threats, multimedia evidence, and the defendant's confession met the legal standard for admissible digital evidence in Saudi courts.

### 4) *Legal Analysis*

Under Saudi Arabia's Anti-Cybercrime Law, extortion, defamation, and threats are criminal offenses. The defendant's actions met the criteria for multiple cybercrime violations.

The defendant's prior extortion case served as an aggravating factor, justifying increased penalties. The Court of Appeal emphasized that repeated cybercrimes required stronger sentencing to deter recurrence.

### 5) *Court Decision Reasoning*

The Criminal Court of Taif based its conviction on:

- Verified digital identity of the defendant
- Clear linkage between the posts and the victim
- The defendant's confession
- Evidence of previous cybercrime behavior

The Court of Appeal upheld the conviction and returned the case twice with instructions to impose a harsher sentence. The court determined that the initial punishment did not sufficiently reflect the aggravated circumstances.

### 6) *Findings and Discussion*

This case demonstrates how Saudi courts handle digital-crime cases involving threats and extortion:

- Authenticated digital evidence is sufficient for conviction.
- Harmful intent and repeated offenses significantly impact sentencing.
- Strict legal responses aim to prevent online exploitation.
- Previous cybercrime records lead to increased penalties.

The case highlights the importance of proper digital-evidence handling and the impact of aggravating factors in legal judgments.

### 7) *Summary of Insights*

The case illustrates how Saudi courts balance forensic evidence, confession, and legal principles to impose enhanced penalties when victims are threatened or targeted online. It reinforces the judiciary's strong stance on deterrence and public protection against digital crimes.

### ***C. Case 3: Data leakage case – NIST Digital forensics dataset***

#### ***1) 1. Case Background***

This forensic case involves a controlled dataset provided for digital-forensics training, focusing on the recovery of Russian UTF-16BE encoded text hidden within a disk image. The simulated scenario represents the theft of a Russian restaurant menu, whose content was concealed inside the image using Unicode encoding. The case is academically significant because it highlights challenges related to multilingual text recovery and encoding-aware forensic techniques.

#### ***2) Incident Description***

It started when an outside party (Mr. Conspirator) had approached Mr. Informant with a request to provide proprietary technical information. The employee copied the confidential materials in the USB device and a CD as well as he tried to transfer the data to personal cloud-storage accounts. Security officers apprehended him at one of the checkpoints within the facility and seized his gadgets before he could walk out of the facility. The gathered evidence was taken to a digital-forensics lab to have all the evidence analyzed.

#### ***3) Evidence Examination***

The data collected during this dataset was comprehensive and technical enough. It included:

- Forensic photos of USB and CD with secret company documents.
- Metadata analysis, which indicates timestamps in agreement with copying and transfer activities.
- Recovery of deleted files, showing evidence of having been deleted.
- Records of the cloud-logs which show the latest uploads made by the employee.
- Checking of hash-values, integrity of extracted data.

Forensic examination has been in line with ISO/IEC 27037 and NIST 800-101 standards and it was also able to prove total authenticity and integrity of the recovered evidence.

#### ***4) Legal Analysis***

The devices were justified to be seized and examined by corporate-security authorization, internal policy and digital-forensic best practices. The actions of the employee were the breach of the unlawful disclosure of the safeguarded information of a proprietary nature, which is a punishable

offense in the majority of cyber-security and intellectual-property laws. The presence of duplicated files, the content of the metadata, and the evidence of logs proved the parts of a planned data breach.

#### ***5) Court Decision rationale***

In case they were presented in court, the evidence would be considered admissible because of:

- Authentication of identity to the storage devices.
- Authenticated and unaltered metadata.
- Definite motive as evidenced by copying files.
- An exfiltration behaviorally compatible documented timeline.
- Quality forensic imaging and integrity protection.

The forensic workflow is as strong as it meets the judicial demands of integrity, authenticity, and relevance.

#### ***6) Findings and Discussion***

The case demonstrates the high evidentiary potential of well-collected and analyzed digital objects in cases of insider-threats. It demonstrates:

- Good correlation of actions, timestamps and ownership of device.
- Good reliability of file recovery logs and logs of cloud activities.
- The relevance of full forensic imaging and hash validation.
- The importance of internal checkpoints in policy violations.

#### ***7) Summary of Insights***

Data Leakage Case is a good model to comprehend the detection and proving of insider threats in the light of detailed forensic investigation. It confirms the importance of metadata, platform records and high levels of forensic standards in proving those who were intent and authenticity during corporate-security inquiries.

### ***D. Case 4: Saudi Digital Extortion Case – Jeddah extortion case***

#### ***1) Case Background***

The case under consideration was heard in the Criminal Court of Jeddah, and it is associated with the accusations of cyber-extortion and harassment of a woman by sending threatening SMS messages. The complainant claimed that the anonymous numbers posted negative words and also threatened to expose the personal photos of the victim. The phone numbers were registered in the name of the defendant, and no technical measures were performed to confirm the authorship and the use of the device. The case is noteworthy

as it shows difficulties that may arise due to the failure of acquiring digital-evidence fully.

## **2) Incident Description**

The victim was given several text messages threatening to publish her personal pictures to her family members. Other messages consisted of insults and suggested violence. Although one of the numbers was found to be associated with the suspect, the accused denied it and gave medical records showing that he was hospitalized and sedated at the same time as the day in question. The investigators failed to recover the digital logs, metadata, and perform a forensic analysis of the device used by the defendant despite the frequent demands put in by the Public Prosecution.

## **3) 3. Digital Evidence Examination**

The electronic evidence presented in the case was low and lacking. Key observations include:

- No forensic copying of the phone/messaging logs of the defendant.
- No metadata is attained, and this does not allow the establishment of the sender identity.
- No device imaging and thus integrity cannot be verified.
- Lacking platform or telecom logs, e.g. IP traces or SIM-activity logs.
- Medical records, in the favor of the defendant, incapacity.

In contrast to the Taif case (sample), the evidence in this case could not be recognized as being of authentication, integrity verification and technical reliability.

## **4) Legal Analysis**

Despite the fact that the case claimed commissions of Saudi Anti-Cybercrime Law (threats, extortion, harm to reputation) there were no digital-forensic procedures so that the main legal aspects could be determined. Since no technical indicator was possible to connect the evidence with the defendant, the legal test of establishing digital acts authorship was not met. The medical evidence by the defendant also served to discredit the prosecution on proving its capability or intent at the time the messages were sent.

## **5) 5. Court Decision Reasoning**

The Criminal Court had dismissed the case on the ground of lack of sufficient evidence because:

- The online messages were not attributed technically.
- The police did not follow required forensic tests.
- There was no chain of custody and integrity check.

- The hospital records of the defendant raised a reasonable doubt.
- The prosecution did not succeed in showing authorship, possession, and intent.

The decision was upheld by the Court of Appeal.

## **6) Findings and Discussion**

The case reflects the extreme reliance of cybercrime prosecution on adequate management of digital-evidence. Key findings include:

- Absence of forensic procedures negate otherwise serious allegations.
- The identity has to be attributed using metadata, device imaging, and telecom logs.
- The medical evidence can play a major role in capability assessment.
- Cases will be thrown out in courts where procedural loopholes are not allowing reliability and authenticity checks.

## **7) Summary of Insights**

The Jeddah case explains why the lack of technical validation makes digital evidence inadmissible even in case the charges involved are severe. It upholds the need to observe strictly the standards of forensic-collection to bring out clarity, reliability and legal attribution in cases of cybercrime.

| Phase                                           | Requirement                     | Verification Questions                                                                                                  | Case 1 | Case 2 | Case 3 | Case 4 |
|-------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|
| Phase 0: Pre-Evidence Validation                | Digital Identity Verification   | Is the account linked to the suspect through IP logs, Device ID, MAC, login history? Is stymetric profiling consistent? | ✓      | ✗      | ✓      | ✗      |
|                                                 | Native Format Requirement       | Is the evidence presented in its original native form (no screenshots)? Are metadata fields intact?                     | ✗      | ✓      | ✓      | ✗      |
|                                                 | Platform Reliability Assessment | Does the platform retain logs? Is the platform vulnerable to manipulation?                                              | ✓      | ✓      | ✓      | ✓      |
| Phase 1: Legal Authorization                    | Lawful Collection               | Was a valid warrant issued? Was collection conducted without violating privacy laws?                                    | ✓      | ✗      | ✓      | ✓      |
|                                                 | Relevance and Scope             | Is the evidence directly related to the investigation?                                                                  | ✓      | ✓      | ✓      | ✓      |
|                                                 | Authorization Audit Trail       | Is there documentation verifying lawful acquisition?                                                                    | ✗      | ✗      | ✓      | ✗      |
| Phase 2: Technical Examination – Pre-Requisite  | Validated Tools                 | Was a tested and certified forensic tool used? Is the tool version documented?                                          | ✓      | ✓      | ✓      | ✗      |
|                                                 | Competent Examiner              | Is the examiner trained and certified?                                                                                  | ✓      | ✓      | ✓      | ✗      |
| Phase 2: Technical Examination – Core           | Chain of Custody (3-Layer)      | Is there complete documentation of collection → transfer → analysis?                                                    | ✓      | ✗      | ✓      | ✗      |
|                                                 | Integrity Verification          | Are hash values (MD5/SHA-256) recorded before and after analysis?                                                       | ✗      | ✓      | ✓      | ✗      |
|                                                 | Forensic Imaging                | Was a bit-for-bit forensic image created?                                                                               | ✓      | ✓      | ✓      | ✗      |
| Phase 2: Technical Examination – Post-Requisite | Forensic Report Completeness    | Does the report describe tools, steps, settings, and findings?                                                          | ✓      | ✓      | ✓      | ✗      |
|                                                 | Expert Testimony Preparedness   | Can the expert defend methods under Daubert scrutiny?                                                                   | ✓      | ✗      | ✓      | ✗      |
| Phase 3: Authenticity & Reliability Check       | Metadata Consistency            | Timestamps, file origin, and authorship consistent?                                                                     | ✓      | ✓      | ✓      | ✗      |
|                                                 | Account Ownership Proof         | Identity linked conclusively to suspect?                                                                                | ✓      | ✗      | ✓      | ✗      |
|                                                 | Tamper Detection                | Any indication of editing or manipulation?                                                                              | ✗      | ✓      | ✗      | ✗      |
|                                                 | Hash Matching                   | Do hash values match across all stages?                                                                                 | ✗      | ✓      | ✓      | ✗      |
|                                                 | Timestamp Logic                 | Are timing sequences coherent with case facts?                                                                          | ✓      | ✓      | ✓      | ✗      |
|                                                 | Contextual Reliability          | Does evidence logically correlate with other data?                                                                      | ✓      | ✓      | ✓      | ✗      |

Table (9)

## IX. CONCLUSION

Digital evidence plays a key role in investigations in legal and cybersecurity fields. However, its acceptance still faces issues due to its unstable nature and problems with proving where it came from. The original model by Antwi-Boasiako and Venter laid important groundwork for lawful permission and technical review, but today's digital-evidence environments need stronger verification methods. The improved framework presented in this paper, E-DEAF, boosts the acceptance of evidence by adding pre-validation, authenticity checks, and organized judicial review. Following forensic-handling standards from NIST, tool-assurance guidance from ISO/IEC 27037, and screenshot-validity suggestions from SWGDE results in more trustworthy workflows. Case studies in Saudi courts, such as the Taif conviction and the dismissal of the Jeddah case, show that successful digital-evidence outcomes rely on a verified digital identity, intact metadata, validated forensic tools, and a well-documented chain of custody. Without these technical protections, evidence may be rejected, even in serious cybercrime cases. Thus, structured admissibility frameworks supported by scientifically validated methods are vital to improve judicial transparency, strengthen expert defensibility, and ensure fair outcomes in digital-age litigation.

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